

Elastic Beanstalk Aws

Overview of Elastic Beanstalk

- Elastic Beanstalk simplifies application deployment by allowing developers to manage applications without worrying about underlying infrastructure complexities⁵.
- The service manages capacity provisioning, load balancing, scaling, and application health monitoring⁵.

Elastic Beanstalk Components

- There are three main components: Application, Application Version, and Environment⁵.
 - **Application:** A logical collection of multiple Elastic Beanstalk components⁵⁵.
 - **Application Version:** The deployable code bundle, such as a Java .war file⁵.
 - **Environment:** A logical collection of resources to run a specific application version, with only one application version running at a time⁵⁵.

Deploying an Environment

- The tutorial covers deploying a simple Elastic Beanstalk environment and an advanced, production-level environment, including high availability configurations⁵⁵.
- Developers can quickly create environments and manage deployment settings⁵.

Deployment Options and Policies

- Various deployment strategies include:
 - **All at Once:** Deploy all instances simultaneously, leading to downtime⁵⁵.
 - **Rolling:** Update instances in batches, which may run at reduced capacity⁵⁵.
 - **Rolling with Additional Batches:** Maintain capacity by adding new instances while updating⁵⁵.
 - **Immutable:** Create a new auto-scaling group for the updated version, ensuring zero downtime⁵⁵.
 - **Blue/Green Deployment:** Redirect traffic between two environments to reduce risk⁵⁵.

Advanced Concepts

- The tutorial discusses advanced topics, including extension files (dot EBS extensions), lifecycle policies, and different environment tiers (web server vs. worker tier)⁵⁵.
- Users can maintain up to 1000 application versions, ensuring that in-use versions cannot be deleted⁵⁵.

Adding a Database

- The session explains how to add a MySQL database instance to an existing Elastic Beanstalk environment⁵⁵.
- Developers are warned that if the environment is terminated, the linked database will also be deleted unless decoupled⁵⁵.

Decoupling a Database

- Decoupling the RDS database from the Elastic Beanstalk environment is crucial to protect data^{ss}.
 - The process involves creating a duplicate environment and linking it to the existing database without losing data^{ss}.
1. What is the purpose of blue/green deployment in Elastic Beanstalk?
 2. What happens to the database when you delete an Elastic Beanstalk environment that is coupled with it?
 3. What are the challenges faced by developers when deploying applications on a cloud platform?
 4. How does auto scaling benefit application deployment in cloud environments?
 5. What is the primary benefit of using Elastic Beanstalk for application deployment?
 6. What are the three main components of Elastic Beanstalk?
 7. What are the three main components of Elastic Beanstalk?
 8. How does Elastic Beanstalk manage application versions and environments?
 9. What is the charge for using Elastic Beanstalk as a service?
 10. What must you do after creating an application in Elastic Beanstalk?
 11. What is the purpose of defining specifications in a web server environment?
 12. Which pre-configured platforms can be used in the AWS Elastic Beanstalk environment?
 13. What is the first step in creating an environment in Elastic Beanstalk?
 14. What type of environment is initially created in the tutorial before moving to a production-level environment?
 15. What are the two types of environments you can deploy in AWS Elastic Beanstalk?
 16. What is the purpose of configuring advanced options when creating a production environment in AWS Elastic Beanstalk?
 17. What is the main disadvantage of the 'all-at-once' deployment strategy?
 18. How does the 'rolling with additional batches' strategy help maintain application capacity during updates?
 19. What is the primary benefit of using an immutable deployment strategy?
 20. How does the immutable deployment strategy affect costs during deployment?
 21. What is Blue/Green Deployment?
 22. What is Blue/Green Deployment?
 23. What happens to an RDS database instance when the associated Elastic Beanstalk environment is terminated?
 24. What is the minimum storage requirement for a MySQL database instance in AWS RDS?
 25. What is the primary reason for decoupling an RDS database from an Elastic Beanstalk environment?